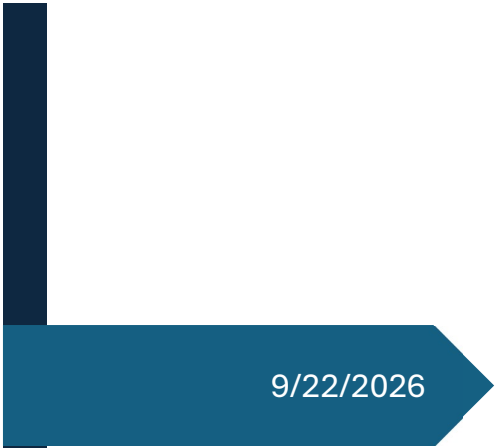
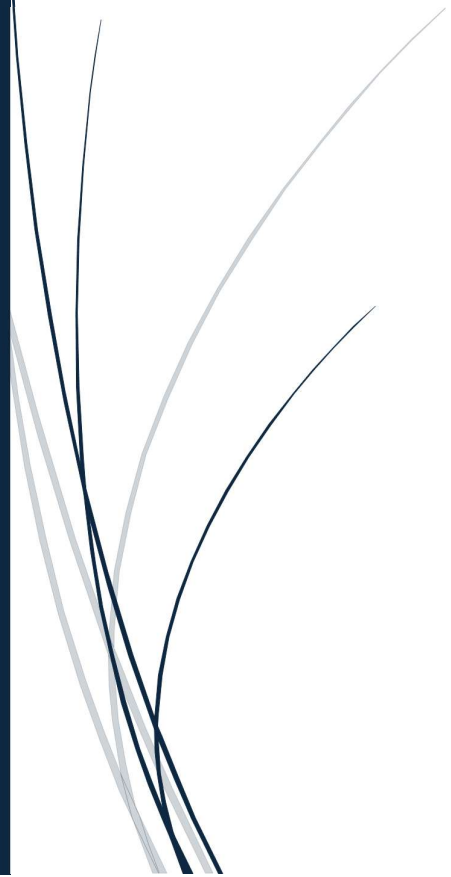


A thick dark blue vertical bar runs down the left side of the page. A dark blue arrow points to the right from the bar, containing the date.

9/22/2026

Programming 2B (PROG6212)

POE Part 1

Several thin, light blue curved lines sweep upwards from the bottom left corner of the page.

Dimpho Motsie
ST10458752

Table of Contents

Section A – Entity Relationship Diagram (ERD) 2

Section B - API Endpoint Plan 3

References 8

Section A – Entity Relationship Diagram (ERD)

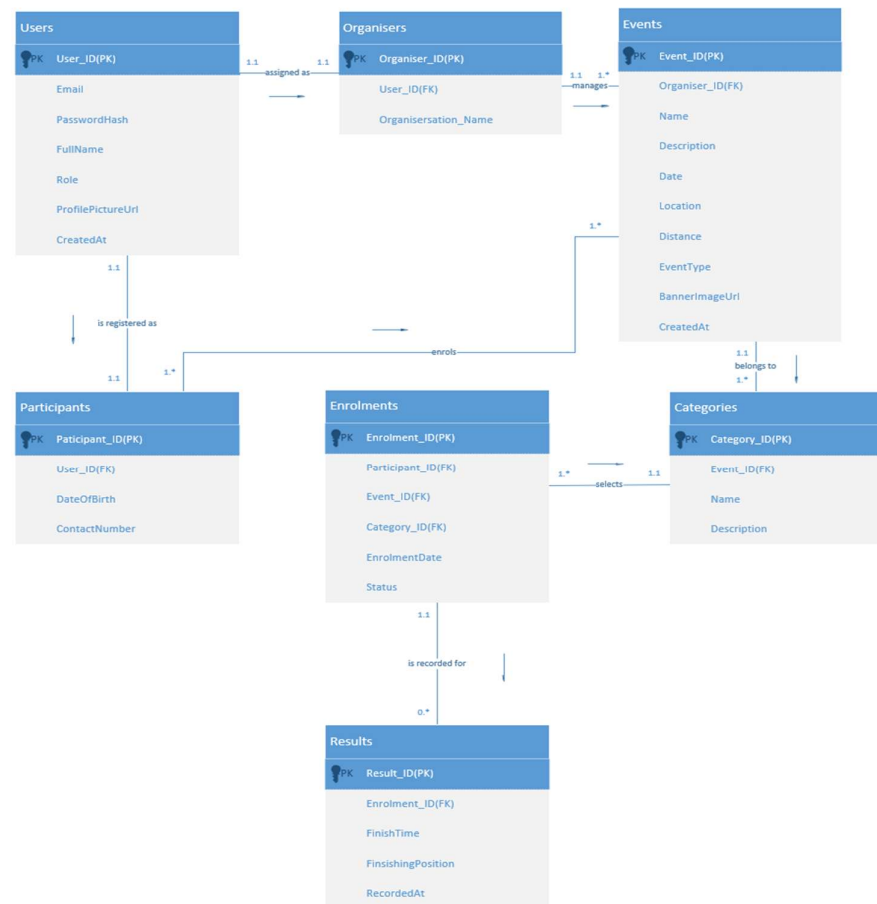


Illustration of an Entity-Relationship Diagram (Microsoft, 2026).

Section B - API Endpoint Plan

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user (Organiser or Participant).	None (Public)	{"email": "string", "password": "string", "fullName": "string", "role": "string", "organisationName": "string? (for Organiser)", "dateOfBirth": "date? (for Participant)", "contactNumber": "string? (for Participant)"} }	201 Created - New user object (without password). 400 Bad Request - Invalid data or email already exists.
POST	/api/auth/login	Authenticates a user and creates a server-side session.	None (Public)	{"email": "string", "password": "string"}	200 OK - User object and role. 401 Unauthorized - Invalid credentials.
POST	/api/auth/logout	Ends the current user's session.	Any (Logged In)	None	204 No Content - Session ended. 401 Unauthorized - User not logged in.

GET	/api/users/profile	Retrieves the profile details of the currently authenticated user.	Any (Logged In)	None	200 OK - User profile object (includes Organiser/Participant details based on role). 401 Unauthorized - User not logged in.
PUT	/api/users/profile	Updates the profile details of the currently authenticated user.	Any (Logged In)	{"fullName": "string", "dateOfBirth": "date?", "contactNumber": "string?", "organisationName": "string?"}	200 OK - Updated user profile object. 400 Bad Request - Invalid data. 401 Unauthorized - User not logged in.
GET	/api/events	Retrieves a list of all events.	Any (Logged In) or None?	None	200 OK - List of event summary objects.
GET	/api/events/{id}	Retrieves detailed information for a specific event.	Any (Logged In) or None?	None	200 OK - Detailed event object including its categories. 404 Not Found - Event does not exist.
POST	/api/events	Creates a new event.	Organiser Only	{"name": "string", "description": "string", "date": "datetime", "location": "string", "distance": "string", "eventType": "string"}	201 Created - New event object. 400 Bad Request - Invalid data. 403 Forbidden - User is not an Organiser.

PUT	/api/events/{id}	Updates an existing event.	Organiser Only	{"name": "string", "description": "string", "date": "datetime", "location": "string", "distance": "string", "eventType": "string"}	200 OK - Updated event object. 400 Bad Request - Invalid data. 403 Forbidden - User is not the owner. 404 Not Found - Event does not exist.
DELETE	/api/events/{id}	Deletes a specific event.	Organiser Only	None	204 No Content - Event deleted. 403 Forbidden - User is not the owner. 404 Not Found - Event does not exist.
GET	/api/events/{id}/categories	Retrieves all categories for a specific event.	Any (Logged In) or None?	None	200 OK - List of category objects for the event. 404 Not Found - Event does not exist.
POST	/api/events/{id}/categories	Adds a new category to a specific event.	Organiser Only	{"name": "string", "description": "string"}	201 Created - New category object. 400 Bad Request - Invalid data. 403 Forbidden - User is not the owner. 404 Not Found - Event does not exist.
DELETE	/api/categories/{id}	Deletes a specific category.	Organiser Only	None	204 No Content - Category deleted. 403 Forbidden - User

					is not the owner of the event. 404 Not Found - Category does not exist.
GET	/api/enrolments	Retrieves all enrolments. For Organisers, returns all enrolments for their events. For Participants, returns their own enrolments.	Any (Logged In)	None	200 OK - List of enrolment objects. 401 Unauthorized - User not logged in.
POST	/api/events/{id}/enrolments	Allows a Participant to enrol in a specific event and category.	Participant Only	{"categoryId": "int"}	201 Created - New enrolment object. 400 Bad Request - Already enrolled or invalid data. 403 Forbidden - User is not a Participant. 404 Not Found - Event or Category does not exist.
GET	/api/events/{id}/enrolments	(Organiser View) Retrieves all enrolments for a specific event.	Organiser Only	None	200 OK - List of enrolments (with participant details) for the event. 403 Forbidden - User is not the owner.

					404 Not Found - Event does not exist.
PUT	/api/enrolments/{id}/status	(Organiser View) Updates the status of an enrolment (e.g., Confirm).	Organiser Only	{"status": "string"}	200 OK - Updated enrolment object. 400 Bad Request - Invalid status. 403 Forbidden - User is not the owner of the event. 404 Not Found - Enrolment does not exist.
POST	/api/enrolments/{id}/results	Allows an Organiser to capture the finish time and position for a specific enrolment after an event.	Organiser Only	{"finishTime": "time", "finishingPosition": "int"}	201 Created - Result object. 400 Bad Request - Result already exists or invalid data. 403 Forbidden - User is not the owner of the event. 404 Not Found - Enrolment does not exist.
GET	/api/participants/results	Allows a Participant to view their own personal race history/results.	Participant Only	None	200 OK - List of result objects (including event, category, finish time, position). 401 Unauthorized - User not logged in.

					403 Forbidden - User is not a Participant.
--	--	--	--	--	--

API Endpoint Plan (Microsoft Learn, 2026a; Microsoft Learn, 2026b; Microsoft Learn, 2026c; Microsoft Learn, 2026d; Microsoft Learn, 2026e; Postman Blog, 2025 & RFC Editor, 2022).

References

Microsoft, 2026. Microsoft Visio. [App]. Available from Microsoft Office Suite. [Accessed 21 September 2026].

Microsoft Learn, 2026. Best practices for RESTful web API design [Online]. Available: <<https://learn.microsoft.com/en-us/azure/architecture/best-practices/api-design>> [Accessed 21 September 2026].

Microsoft Learn, 2026. Routing to controller actions in ASP.NET Core [Online]. Available: <<https://learn.microsoft.com/en-us/aspnet/core/mvc/controllers/routing?view=aspnetcore-10.0>> [Accessed 21 September 2026].

Microsoft Learn, 2026. Controller action return types in ASP.NET Core web API [Online]. Available: <<https://learn.microsoft.com/en-us/aspnet/core/web-api/action-return-types?view=aspnetcore-10.0>> [Accessed 21 September 2026].

Microsoft Learn, 2026. Role-based authorization in ASP.NET Core [Online]. Available: <<https://learn.microsoft.com/en-us/aspnet/core/security/authorization/roles?preserve-view=true&view=aspnetcore-5.0>> [Accessed 21 September 2026].

Microsoft Learn, 2026. Model Validation in ASP.NET Web API [Online]. Available: <<https://learn.microsoft.com/en-us/aspnet/web-api/overview/formats-and-model-binding/model-validation-in-aspnet-web-api>> [Accessed 21 September 2026].

Postman Blog, 2025. REST API Best Practices: A developer's guide to building reliable APIs. [Online]. Available: <<https://blog.postman.com/rest-api-best-practices/>> [Accessed 21 September 2026].

RFC Editor, 2022. HTTP Semantics (RFC 9110) [Online]. Available: <<https://www.rfc-editor.org/rfc/rfc9110.html>> [Accessed 21 September 2026].